Sunburn (Desert / Canyon / Long-Range Fire & Air Superiority)

Executive Summary

Sunburn is a blitzkrieg **desert-canyon battlefield** built for **long-range firepower**, **air superiority**, **high-altitude reconnaissance**, **and autonomous maneuverin across open and vertical terrain**. The environment features sandstone canyons. wind-carved ridges. dry riverbeds. elevated plateaus. abandoned mining sites. and sparse desert settlements. The mao-emphasizes Obsidian Protocol's autonomous lone-ranee tarseting lastic. air-eround coordination. sensor-fusion under heat distortion. and logistics under extreme desert conditions. In AR Nreal Lights'Air1. the mao anchors at ~4–5 m viewing distance. coverine a 10×10 m ohvsical area representing ~1 km² of canvon-desert terrain. Kev features include canvon siehtlines. Neh-eround air corridors. lone-ranee artillerv zones. and heat-driven sensor deeradation.

Design Goals & Gameplay

Long-Range Desert Firepower
Sunburn prioritizes snipers, artillery, missile vehicles, and long-range drones. Open desert flats and canyon ridges create extreme sightlines.
Air Superiority & High-Altitude Recon
Aircraft, VTOLs, and drones dominate the vertical dimension. Air corridors between canyon walls define recon, interception, and strike patterns.
Canyon Maneuvering & Vertical Combat
Ground units navigate narrow canyon passes, elevated ridges, and dry riverbeds. Verticality creates ambush zones and long-range overwatch positions.
Autonomy & Heat-Aware Pathfinding
Autonomous units choose routes based on heat shimmer, elevation, cover scarcity, and sensor clarity. Units avoid exposed flats during peak heat unless necessary.
Sensor & Information Warfare
Heat distortion, dust storms, mirage effects, and thermal plumes degrade sensors. Relay towers on ridges restore clarity. Destroying relays creates blind pockets across the canyon network.
Logistics & Resources
Fuel depots, mining stations, and desert outposts serve as resupply nodes. Convoys must navigate exposed roads and avoid ambush-prone canyon passes.
Deployment & Progress
A fixed deployment budget (~10,000 points) forces players to choose between long-range artillery, air dominance, or canyon-adapted armor. Blue deploys from the southwest dry-river staging zone; Red deploys from the northeast ridge-top command.

Spatial Constraints (Nreal AR)

Device Capabilities
Nreal Light/Air; ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.
Playable Area & Scaling
10×10 m physical area = 1 km² canyon-desert zone (1:100 scale). Designed for tabletop AR.
Tracking & Occlusion
Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System
1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.
Key Locations
Blue HQ Grove•Gear Staging Zone1 — (0,0). 150×150 m Cargo crates. armored vehicles. drone pads.
Red HQ Hidden•Foo Command1 — (1000,1000). 150×150 m Elevated bunker with radar mast. AA emplacements. and lone-ranee sensors.
Central Canyon Basin — (500,450). 300×300 m Wide canyon floor. lone sightlines. heat shimmer. artillery duels.
Mining Station Depot — (250,800). 120×120 m Abandoned machinerv. fuel tanks. interior flankine routes.
Ridge Network — (800,300). 200×200 m High-eround overwatch. sniper nests. air-corridor control.
Canyon Pass Corridor — (0,450 → 400,900) Primary chokepoint: narrow. exposed. ideal for ambushes.
Desert Road Route — (800,0 → 650,350) Safer but predictable: vulnerable to overwatch from ridges.
Ridge Relay A — (650,550). +240 m Lone-ranee sensor vantage point.
Ridge Relay B — (150,650). +260 m Comms relay hub and air-control node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Canyon Pass Corridor
Major chokepoint; controlling it determines ground movement.
Ridge Network
Extreme long-range sightlines; ideal for snipers and air-ground coordination.
Canyon Basin
Long-range artillery and missile engagements; heat shimmer affects accuracy.
Desert Road Route
Predictable but safe; vulnerable to ridge overwatch.
Sightlines
Ridge Relay A provides long-range sensor coverage (~600 m). Heat distortion and dust storms create fog-of-war pockets.

Deployment Zones

Blue Deployment SW dry-river staging + armor
Red Deployment NE ridge-top command + AA emplacements
Neutral Forward Zone Outpost at (500,950) for drone ferry

Assets & Environment

Terrain
Canyons, ridges, desert flats, dry riverbeds, mining stations.
Vegetation
Sparse shrubs, desert grass, small trees.
Buildings & Props
Military: AA guns, radar dishes, bunkers. Industrial: Mining equipment, generators, pipelines. Logistics: Trucks, crates, fuel drums, drones.
Effects
Dust storms, heat shimmer, mirage distortion, wind gusts.
Audio
Wind, distant machinery, radio chatter, aircraft.

LOD & Budget

Asset Cate	Count	Tris/Item	Total Tris
Canyon/R	3	200k	600k
Industrial	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget
2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins
Unreal AREmulator + Nreal SDK (XREAL). ARKit/ARCore enabled.
Rendering
Forward renderer, stationary lights, baked GI, aggressive culling.
Anchors
Single spatial anchor for entire map.
Interaction
Phone pointer for pan/zoom; HUD as screen-space widgets.
Multiplayer
Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones
1. Prototype Layout — Month 1 Block-out canyon basin. ridges. mining station. AR anchor test.
2. Core Systems — Month 2 NavMesh. lone-ranee autonomv behaviors. air-eround coordination.
3. AR Interaction — Month 3 Anchors. plane detection. basic interactivity.
4. Content Fill — Month 4–5 Final models. textures. lighting. audio.
5. Gameplay & Polish — Month 6–7 Objectives. balance. VFX (dust storms. heat shimmer).
6. Testing & QA — Month 8 Multiplaver stress test. sensor validation.

Testing Plan

Autonomy
Verify long-range targeting, air-ground coordination, heat-aware navigation.
Comms/Sensors
Test relay destruction, fog-of-war updates, LOS checks under heat distortion.
Performance
≥60 Hz AR render on target device.
QA Checklist
Navigation, long-range logic, deployment budget enforcement, anchor drift, sensor accuracy.